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Roll No.

Degree: ...B.TECH - ECAM... Semester: 3rd...

END-SEMESTER THEORY EXAMINATION, NOV-DEC,2024

Course Code: EAEPC305

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 03 Hours

Max. Marks: 50

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	1. What is the maximum possible count with 3,4,8,16 bit binary numbers?	2	CO1
	2. Convert the following: a. $(10101)_2$ to Hexadecimal b. $(734)_8$ to Decimal	3	
1b	Explain the importance of parity bit and the significance of hamming code. Derive the hamming code for the given 4-bit data using even parity: $(1010)_2$	5	CO1
1c	Design a CMOS circuit to implement $\bar{F} = (\bar{A}\bar{B}D + B\bar{C})$	5	CO1.2
Q 2	Attempt any 2 parts of the following		
2a	Design an Excess-3 to BCD code converter, using NAND-NAND logic. (All outputs must be <i>high</i> , for any invalid inputs).	5	CO2
2b	Implement $F(w, x, y, z) = \sum \text{min}(1, 2, 5, 6, 9, 12, 13, 14)$ using a 4:1 Multiplexer.	5	CO2
2c	Explain the advantage of carry-look ahead generator and draw the logic diagram of 4-bit generator circuit.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	Differentiate between Latch and Flip-Flop. Draw the logic diagram of JK-Flip-Flop using NAND gates. Explain with the help of characteristic table and derive its characteristic equation.	5	CO3

3b	Design a synchronous 3-bit Down counter, using D Flip-flops.	5	CO3
3c	Draw an asynchronous Mod-10 up counter and explain in detail with the help of timing waveforms.	5	CO3
Q 4	Attempt any 2 parts of the following		
4a	Explain the importance of hardware description language and write notes on the data objects- signals, variables and constants used in VHDL.	5	CO4
4b	Draw logic diagram and write code for VHDL module of 3 to 8 decoder.	5	CO4
4c	Draw logic diagram and write code for VHDL module of Mod-12 counter.	5	CO4
Q 5	Attempt any 2 parts of the following		
5a	Draw the Mealey model and design FSM to detect the sequence (1101) in the incoming serial binary data.	5	CO5
5b	Explain full adder implementation using PLA.	5	CO5
5c	Write short notes on (i) Programmable Logic Devices and (ii) FPGA	5	CO5